



Enhancing SBR/XNBR blend nanocomposites through HNTs reinforcement

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Abstract

This study explores the influence of halloysite nanotubes (HNTs) on the performance characteristics of styrene-butadiene rubber (SBR) and carboxylated acrylonitrile butadiene rubber (XNBR) composites. HNTs were incorporated into the SBR/XNBR matrix at varying loadings (2, 4, 6, 8, and 10 parts per hundred rubber, phr) to assess their effects on curing behavior, mechanical strength, abrasion resistance, and solvent resistance. The incorporation of HNTs markedly accelerated the vulcanization process, as indicated by reduced scorch and optimum cure times, along with increased torque values during rheometric analysis. Among the tested formulations, the composite containing 6 phr HNTs demonstrated the most notable improvements, with tensile strength increasing by 160%, elongation at break by 24%, stress at 100% elongation by 97%, and tear strength by 91%, relative to the unfilled SBR/XNBR blend. In addition to enhanced mechanical performance, the inclusion of HNTs also improved the composites' hardness, abrasion resistance, and compression set behavior, indicating increased crosslinking density and filler-matrix interaction. These findings suggest that HNTs serve as effective multifunctional nanofillers, capable of simultaneously reinforcing and optimizing the cure characteristics of SBR/XNBR rubber blends, with 6 phr identified as the optimal loading for balanced property enhancement.

Keywords SBR/XNBR · HNTs · Cure characteristics · Mechanical properties · Swelling resistance

Introduction

Rubber nanocomposites reinforced with layered silicates represent a novel class of advanced materials with promising applications in diverse sectors, including aerospace, tire manufacturing, automotive components, and high-performance engineering systems [1–4]. The incorporation of nanoclays has attracted considerable interest due to their remarkable ability to enhance the mechanical properties of elastomers—such as tensile strength and elasticity—even at relatively low filler concentrations [5–8]. These improvements are primarily attributed to the distinctive structural characteristics of layered silicates, notably their high aspect ratio and extensive surface area, which facilitate effective stress transfer within the polymer matrix [9–11]. However, the success of such reinforcement is critically dependent on achieving uniform dispersion of the nanoclay within the rubber matrix, as agglomeration can significantly hinder performance benefits [12]. Recent research has consistently demonstrated that the strategic integration of layered silicate fillers leads to notable enhancements in both mechanical and thermal properties of elastomeric composites, reaffirming

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their potential in developing next-generation rubber materials with superior functionality [13].

Traditional rubber composites have commonly employed inorganic fillers such as mica, fly ash, carbon black, and silica to enhance performance [14–18]. However, the effective dispersion of these micron-sized fillers within the rubber matrix poses significant challenges, primarily due to their limited compatibility with the polymer and weak interfacial adhesion [19–21]. To overcome these limitations, nanometer-scale inorganic fillers have emerged as a promising alternative, offering enhanced performance over conventional fillers [22–26]. Nanomaterials such as carbon nanotubes, nanocalcium carbonate, nanosilica, and layered silicates have garnered significant interest as reinforcing agents in rubber systems. Their exceptional surface area, aspect ratio, and mechanical properties enable better interaction with the rubber matrix, resulting in composites with improved mechanical, thermal, and dynamic performance [27–33].

Naturally occurring halloysite nanotubes (HNTs) have garnered significant interest due to their unique structural and functional attributes. Unlike carbon nanotubes (CNTs), HNTs are non-toxic and biocompatible, making them particularly attractive for environmentally friendly and biomedical applications. These aluminosilicate nanotubes typically range from 1 to 3 μm in length and 30 to 70 nm in diameter, with the general chemical formula $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4 \cdot n\text{H}_2\text{O}$ [34–38]. HNTs find widespread use in various fields such as ceramics, cosmetics, drug delivery, and catalysis, owing to their high surface area, tubular morphology, and functional versatility [39–47]. Their crystalline structure and the relatively low density of hydroxyl groups contribute to excellent dispersibility within rubber matrices, overcoming a common challenge associated with traditional fillers [48, 49]. Moreover, the presence of surface siloxane and hydroxyl groups facilitates hydrogen bonding with rubber chains, promoting strong interfacial interactions. This enhanced adhesion at the filler–matrix interface plays a pivotal role in improving the mechanical, thermal, and barrier properties of rubber nanocomposites [50–53].

Rubber blends are widely utilized across various industries, with tire manufacturing being a prominent application area [54]. While conventional rubber blending remains prevalent, the demand for specialized rubbers tailored to meet specific performance criteria is steadily increasing [55]. Styrene-butadiene rubber (SBR), a non-polar synthetic elastomer, is known for its excellent strength, elasticity, and resilience. However, its limited resistance to oils and chemicals poses challenges in harsh operating conditions. In contrast, carboxylated acrylonitrile butadiene rubber (XNBR), a polar elastomer, offers superior resistance to hydrocarbons, oils, oxidative aging, and abrasion. By blending SBR with XNBR, manufacturers can harness the advantageous

properties of both materials, achieving a balanced combination of mechanical performance, chemical resistance, and durability. This approach not only enhances overall material functionality but also ensures cost-effectiveness for industrial applications [56].

This study presents a novel investigation into the reinforcing role of HNTs in SBR and XNBR composites. Unlike prior research that largely focuses on single-polymer systems, this work systematically explores the impact of HNTs on the vulcanization behavior, mechanical performance, abrasion resistance, compression set, swelling resistance, and crosslinking density of SBR/XNBR blends. A distinctive feature of this study is its approach to mitigating the inherent phase incompatibility between SBR and XNBR by employing HNTs not only as reinforcing fillers but also as compatibilizing agents. By incorporating varying loadings of HNTs, the study evaluates how nanofiller dispersion influences the composite's microstructure and crosslinking efficiency. This provides valuable insights into designing cost-effective, high-performance elastomeric materials. The choice of SBR, XNBR, and HNTs reflects a practical and scalable material system, with the potential to yield durable rubber composites tailored for demanding applications. The findings of this research support the advancement of next-generation rubber materials suitable for automotive, aerospace, and industrial uses, where enhanced mechanical strength and environmental resistance are essential.

Experimental

Materials

In this study, Styrene-Butadiene Rubber (SBR1502) containing 23.5% styrene and Carboxylated Acrylonitrile Butadiene Rubber (XNBR 6721) with a solid content of 43% were used as the base elastomers. A conventional vulcanization system was employed, consisting of N-isopropyl-N'-phenyl-p-phenylenediamine (antioxidant 4010NA), zinc oxide, N-cyclohexyl-2-benzothiazole sulfenamide (accelerator CZ), stearic acid, and sulfur. HNTs, supplied by Sigma-Aldrich (Puducherry), were used as the reinforcing nanofiller. These HNTs possess a cylindrical tubular morphology, with a specific gravity of 2.54 g/cm^3 , lengths ranging from 1 to 3 μm , diameters between 30 and 70 nm, and a surface area of 65 m^2/g .

Preparation of SBR/XNBR/HNTs nanocomposites

HNTs were first dispersed in a blended latex mixture of SBR and XNBR using magnetic stirring for two hours to ensure uniform distribution. The resulting dispersion was then

coagulated by introducing it into an aqueous sodium chloride solution acidified with sulfuric acid. The coagulated mass was thoroughly washed multiple times with deionized water to remove impurities and subsequently dried at 50 °C until a constant weight was obtained. The dried rubber blend was processed on a two-roll mill, during which the vulcanization ingredients were added. The formulation included SBR (95 phr), XNBR (5 phr), antioxidant 4010NA (2 phr), zinc oxide (2 phr), stearic acid (2.4 phr), accelerator CZ (2.2 phr), sulfur (1.5 phr), and varying HNT loadings ranging from 0 to 10 phr [57]. The prepared compound was then vulcanized into nanocomposite sheets using a compression molding process at 160 °C under a pressure of 15 MPa for the determined optimum cure time [58].

Characterization

The curing characteristics of the unvulcanized rubber composites were assessed using an oscillating disc rheometer under the conditions specified in ASTM D-2084, operating at 160 °C, 100 cycles per minute, with a 0.5° oscillation arc [59]. Mechanical properties, including tensile and tear strength, were evaluated at room temperature in accordance with ASTM D-412 [60] and ASTM D-624 [61], respectively, using a universal testing machine (Dak System Inc., Series 7200) set at a crosshead speed of 500 mm/min. Additional performance metrics such as hardness, rebound resilience, abrasion resistance, and compression set were measured following ASTM D-2240 [62], ASTM D-2632 [63], ASTM D-5963 [64], and ASTM D-395 [65], respectively. Hardness was measured using a conventional Shore A dial-type durometer by pressing the indenter onto the sample surface under standard conditions. Readings were taken at 5–8 different locations, and the most consistently repeatable value was reported. Swelling resistance was investigated in various solvents at ambient temperature, in line with ASTM D-471 [66]. All reported results represent the mean values derived from five replicate specimens for each mechanical test conducted.

The vulcanized nanocomposite rubber samples were cut into square specimens measuring $25 \times 25 \times 2$ mm³, with care taken to include the cured edges in each piece to maintain structural integrity during testing. Swelling experiments were conducted using 100 mL of toluene, a solvent selected for its compatibility with the rubber matrix. To minimize the influence of light on the swelling behavior, the beaker containing the samples was placed in a dark room maintained at a constant temperature of 23 °C. The samples were immersed in toluene for a period of three days to allow sufficient time for the solvent to diffuse into the rubber matrix and reach equilibrium. To maintain solvent effectiveness and prevent saturation, the toluene was replaced every 12 h,

ensuring continuous exposure to fresh solvent throughout the swelling period. This method enhances the reliability and consistency of the swelling measurements. The crosslink density (ν) of the rubber composites was subsequently calculated using the Flory-Rehner equation, as outlined in previous studies [67–70].

$$M_c = \frac{-\rho_p V_s V_r^{1/3}}{\ln(1 - V_r) + V_r + \chi V_r^2} \quad (1)$$

$$V_r = \frac{1}{1 + Q_m} \quad (2)$$

$$\nu = \frac{1}{2M_c} \quad (3)$$

This equation includes the following terms: ‘ ν ’ denotes the crosslink density; ‘ M_c ’ is the molar mass of polymer chains between crosslinks; ‘ V_r ’ represents the volume fraction of elastomer in the solvent-swollen filled composite; ‘ Q_m ’ indicates the weight swelling ratio of the rubber nanocomposites containing nanofillers in the solvent; ‘ ρ_p ’ stands for the polymer density; ‘ χ ’ is the polymer–solvent interaction parameter, valued at 0.38; and ‘ V_s ’ corresponds to the molar volume of toluene.

The fracture surface morphology of the rubber composite samples was examined using field emission scanning electron microscopy (FESEM, Hitachi S-4160) operating at an accelerating voltage of 3 kV. Prior to imaging, the sample surfaces were coated with a thin layer of gold to enhance conductivity and image quality [71].

Results and discussions

Cure characteristics of the SBR/XNBR/HNTs nanocomposites

Table 1 shows the minimum (M_L) and maximum torque (M_H), delta torque (ΔM), scorch time (t_{s2}), optimum cure time (t_{90}), and cure rate index (CRI) for the SBR/XNBR/HNTs nanocomposites. The M_L represents the initial viscosity of the rubber compound during the curing process. In the SBR/XNBR/HNTs nanocomposites, the increase in M_L with higher HNT content is attributed to the formation of hydrogen bonds and physical entanglements between the polymer chains and HNTs. The tubular structure and high surface area of HNTs enhance interfacial adhesion, restricting chain mobility and leading to higher M_L values [72]. These interactions hinder the movement of polymer chains, making the compound more resistant to deformation. As HNTs content rises, nanofillers are more uniformly

Table 1 Cure properties of SBR/XNBR/HNTs nanocomposites

Sample code	Minimum torque (M _L)	Maximum torque (M _H)	Delta torque (ΔM)	Scorch time (t _{s2})	Optimum cure time (t ₉₀)	Cure rate index (CRI)
	dN m	dN.m	dN m	min	min	min ⁻¹
H ₀	1.06	5.80	4.74	2.15	6.54	22.78
H ₂	1.09	5.92	4.83	2.02	6.38	22.94
H ₄	1.12	6.18	5.06	1.94	6.25	23.20
H ₆	1.15	6.38	5.23	1.78	5.81	24.81
H ₈	1.17	6.45	5.28	1.74	5.74	25.00
H ₁₀	1.19	6.66	5.47	1.68	5.65	25.19

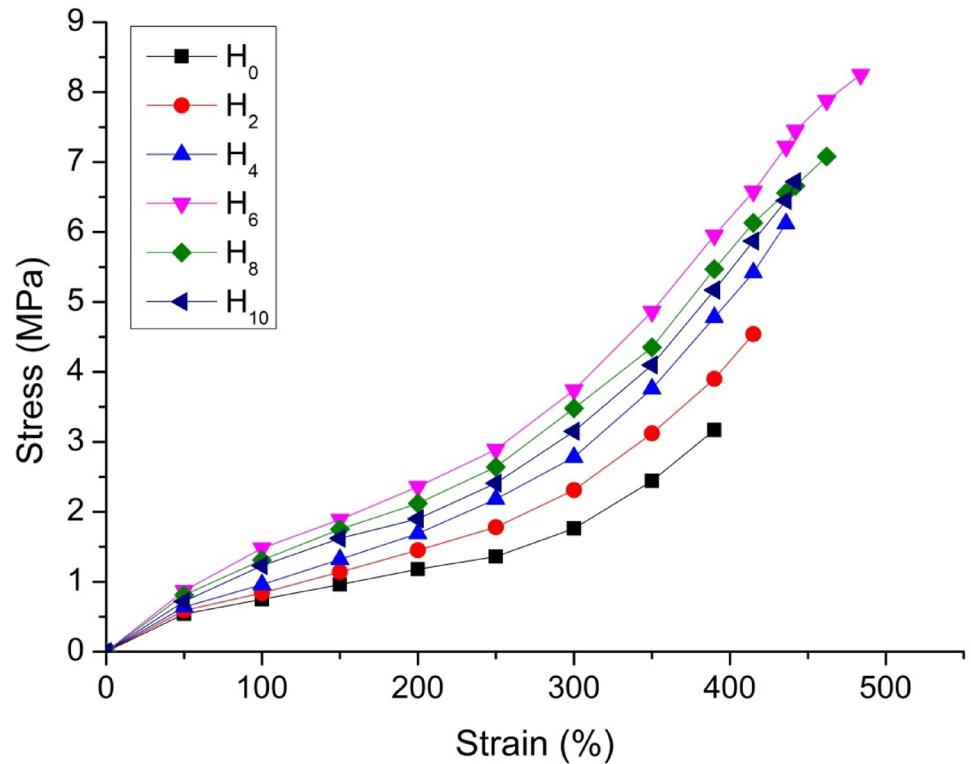
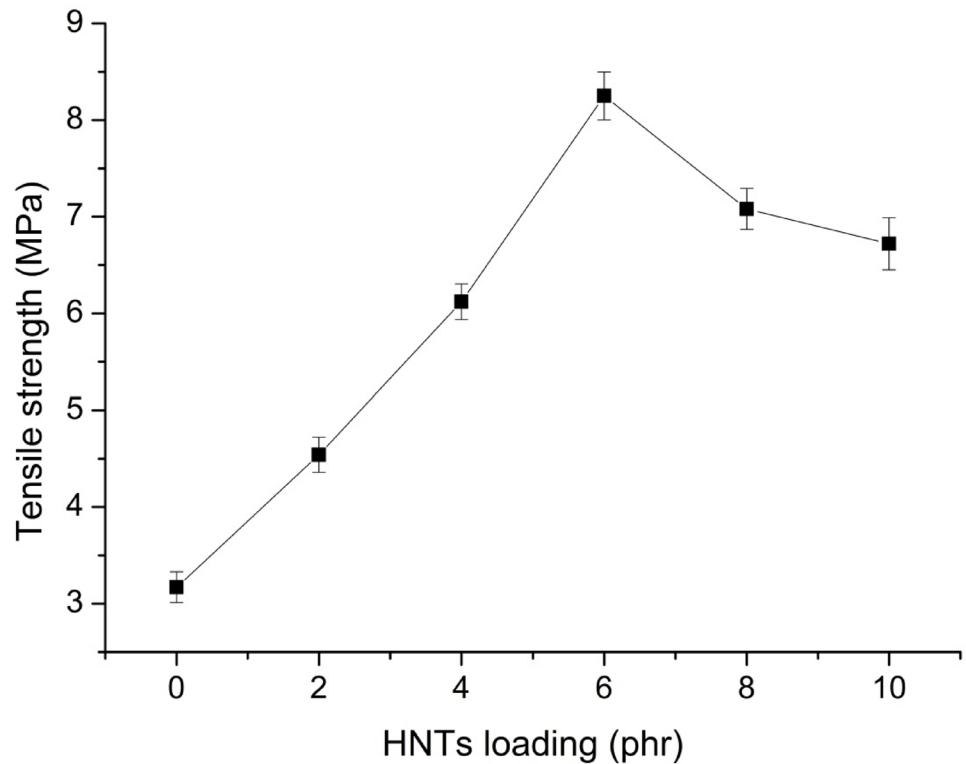
distributed within the matrix, creating a denser network that further increases viscosity. Additionally, the restricted polymer mobility caused by the denser structure contributes to the higher M_L values observed in the nanocomposites [73]. The M_H represents the stiffness and crosslinking density (CLD) of the fully cured material. M_H increases as more HNTs are added to the SBR/XNBR nanocomposites. HNTs serve as additional crosslinking sites, resulting in a stronger and more interconnected network within the rubber matrix, which leads to a stiffer and more rigid material. Moreover, the uniform dispersion of HNTs throughout the polymer blend creates a homogeneous structure that further enhances the overall stiffness of the composite. The reduced free volume caused by the presence of nanofillers also contributes to the increased rigidity, as reflected in higher M_H values [74]. ΔM , the difference between M_H and M_L, represents the extent of crosslinking and the material's final rigidity. As the HNTs content increases, the number of crosslinking sites also increases, resulting in a more rigid composite. The improved dispersion of HNTs within the polymer matrix boosts the uniformity and mechanical strength of the material, leading to a greater ΔM . Additionally, the restricted mobility of polymer chains and the formation of a denser network around the nanofillers further contribute to higher delta torque, reflecting the overall stiffness of the composite as more HNTs are incorporated [75].

Scorch time (t_{s2}) represents the time required for the curing process to begin. As the HNTs content increases in SBR/XNBR/HNTs nanocomposites, t_{s2} decreases. This reduction occurs because HNTs act as activators, promoting the breakdown of curing agents and speeding up the initiation of crosslinking reactions. Additionally, the high surface area of HNTs provides more sites for chemical reactions to occur, promoting faster crosslinking initiation. Improved heat distribution from HNTs also contributes to faster curing, resulting in a shorter t_{s2} [76]. The t₉₀ is the time required for the nanocomposite to reach 90% of its crosslinking and attain its maximum properties. In SBR/XNBR/HNTs nanocomposites, t₉₀ decreases with increased HNTs content. The presence of HNTs provides additional crosslinking

sites, facilitating a faster curing process. Furthermore, the improved thermal conductivity of the composite ensures efficient heat distribution, accelerating the curing process. The higher surface area of the HNTs also allows more efficient interactions with the curing agents, leading to a quicker crosslinking process and reduced t₉₀ [77]. The CRI measures how quickly a rubber compound undergoes curing after crosslinking begins. As the HNTs content increases, CRI also rises. This increase is attributed to the additional crosslinking sites provided by HNTs, which accelerates the curing process. The improved thermal conductivity of the nanocomposite, due to the presence of HNTs, ensures efficient heat distribution throughout the material, speeding up the curing reaction. Furthermore, the catalytic properties of HNTs promote faster decomposition of curing agents, contributing to a higher cure rate and increased CRI values [78].

Mechanical properties of the SBR/XNBR/HNTs nanocomposites

Figure 1 presents the stress–strain curves of the SBR/XNBR/HNTs nanocomposites. From this graph, tensile strength (TS), elongation at break (EB), and stress at 100% elongation (M100) were determined and are shown in Figs. 2, 3, and 4, respectively. The TS of the nanocomposites increases significantly with the addition of HNTs, reaching a peak at 6 phr, where it exhibits approximately a 160% enhancement compared to the unfilled blend. This marked improvement is attributed to the high aspect ratio and tubular morphology of the HNTs, which allow for effective mechanical interlocking and stress transfer within the polymer matrix. However, such reinforcement efficiency is highly dependent on the dispersion quality of the nanofillers. At 6 phr, the conclusion of uniform dispersion is supported by multiple observations. First, the sharp increase in TS indicates efficient load transfer, which can only be achieved when HNTs are homogeneously distributed within the rubber matrix [24, 79, 80]. Second, FESEM images of the tensile fracture surfaces (Fig. 11) show no evidence of agglomeration, phase separation, or localized filler-rich domains; instead,

Fig. 1 The stress–strain curves of SBR/XNBR/HNTs nanocomposites**Fig. 2** The tensile strength of SBR/XNBR/HNTs nanocomposites

the micrographs reveal a continuous matrix with finely dispersed HNT structures, consistent with improved mechanical integrity and stress distribution. Third, the CLD data presented in Fig. 10 demonstrate a notable increase at 6

phr, suggesting that HNTs are well integrated into the cross-linked network through interactions between their hydroxyl groups and the polar functional groups of the rubber chains (nitrile and carboxylic groups in XNBR). Together, the

Fig. 3 The elongation at break of SBR/XNBR/HNTs nanocomposites

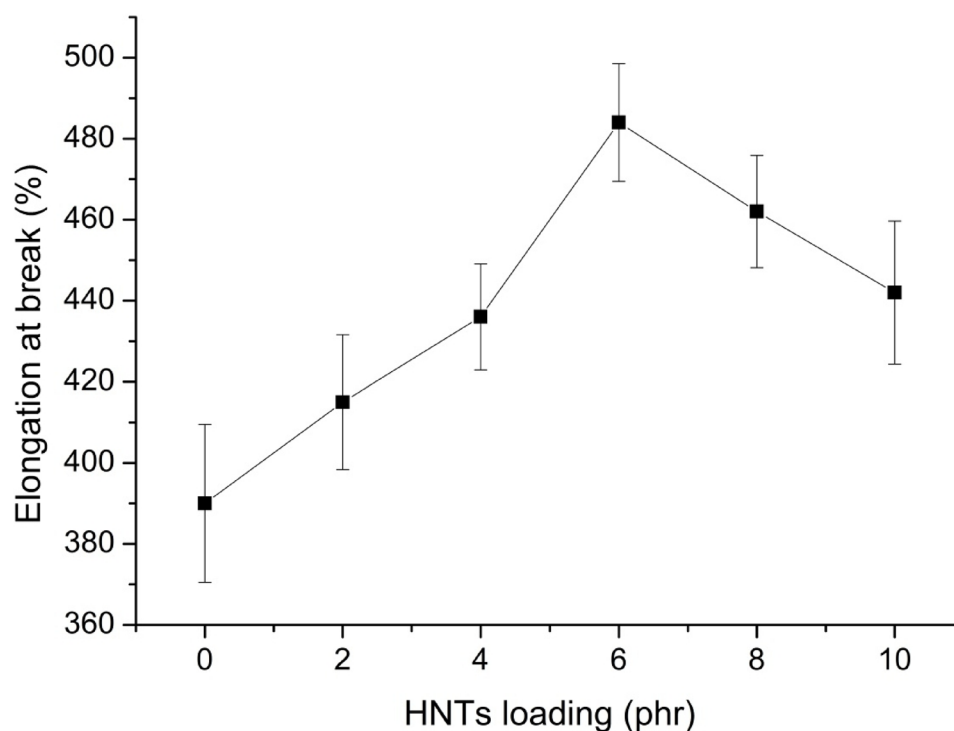
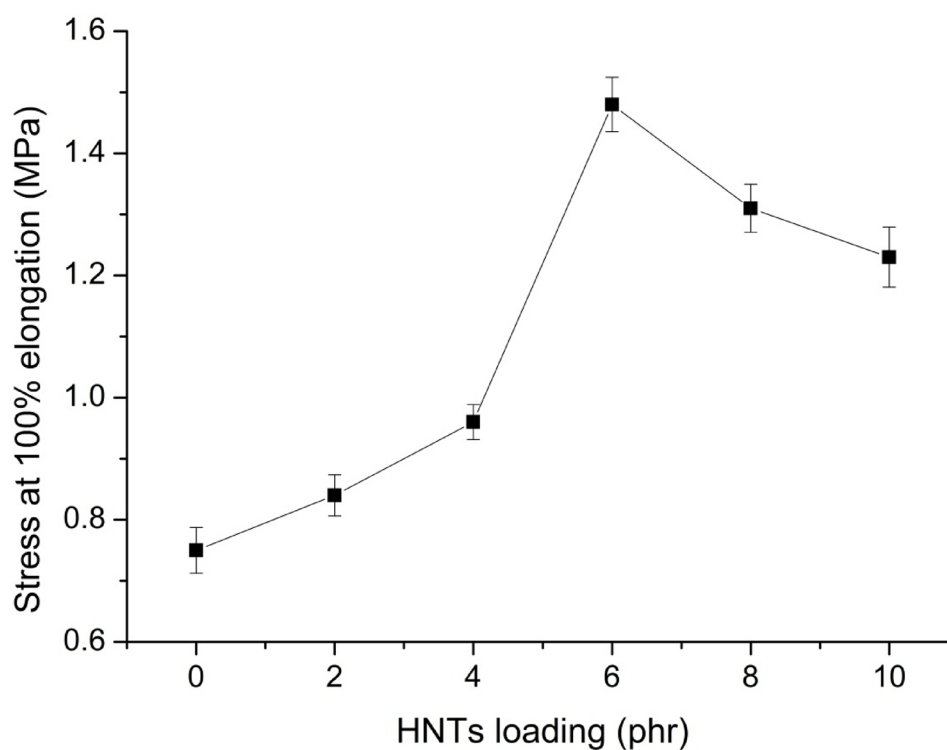


Fig. 4 The stress at 100% elongation of SBR/XNBR/HNTs nanocomposites



reinforcement behavior—marked TS improvement, morphological evidence, and increased CLD—demonstrates minimized filler–filler interaction and maximized filler–polymer interaction. Beyond 6 phr, TS begins to decline, attributed to filler agglomeration that introduces stress concentration sites and disrupts network uniformity [81–83].

Therefore, the combination of mechanical, morphological, and crosslinking data strongly supports the conclusion that the HNTs are uniformly dispersed at 6 phr, resulting in optimal mechanical performance of the SBR/XNBR/HNT nanocomposites.

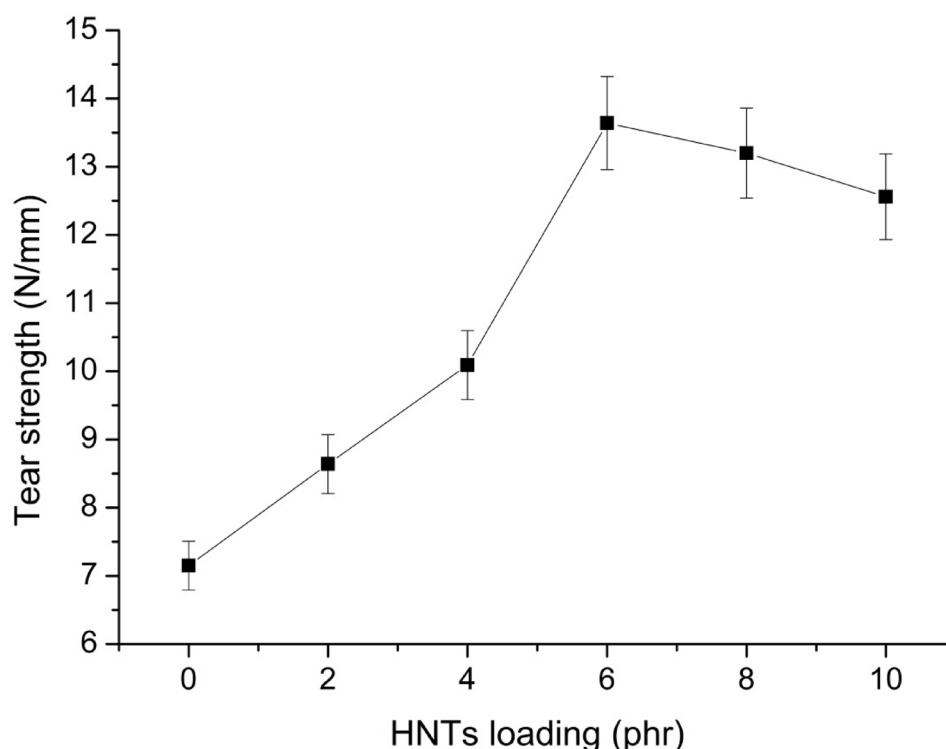
Figure 3 illustrates the EB of SBR/XNBR/HNTs nanocomposites. EB increases with HNTs content, peaking at 6 phr, and then decreases with further increases in HNTs. Up to this optimal loading, HNTs act as effective reinforcing agents, improving stress transfer between the SBR/XNBR matrix and the nanofillers. This reinforcement enhances the composite's ability to withstand deformation, leading to a higher EB. The proper dispersion of HNTs at the optimal loading improves CLD, reinforcing the polymer matrix and contributing to greater toughness and ductility. Moderate HNTs content enhances the interaction between polymer chains, allowing for more elongation before failure. The increased CLD helps maintain composite integrity under tensile stress, enabling greater elongation [84, 85]. However, beyond the optimal loading, an excess of HNTs can lead to agglomeration, resulting in stress concentration points that diminish the composite's elongation capacity. This clustering weakens interfacial bonding, making the composite more brittle and decreasing its EB. High filler content can also restrict polymer chain mobility, further reducing elongation and increasing the likelihood of brittle fracture. The EB of SBR/XNBR/HNTs nanocomposites is 24% higher than that of SBR/XNBR blends. The incorporation of HNTs boosts CLD, fostering improved crosslinking between polymer chains and enhancing the mechanical properties [86]. This increased CLD improves the material's elongation capabilities, allowing the composite to stretch further before breaking.

Figure 4 shows the M100 values of SBR/XNBR/HNTs nanocomposites. M100 increases with HNTs content up to 6 phr, after which it decreases. The optimal dispersion and reinforcement of HNTs occur up to this level, enhancing the composite's stiffness and TS, thereby increasing the stress required for 100% elongation. This increase in stress is due to enhanced load transfer from the polymer matrix to the rigid HNTs. At lower loadings, optimal dispersion of HNTs maximizes the interaction between the nanofillers and polymer chains, improving the composite's resistance to deformation under tensile stress and resulting in higher M100 values. Up to 6 phr, HNTs promote an increase in CLD, which enhances the polymer network structure, providing greater resistance to deformation. The controlled restriction of polymer chain mobility at this stage further contributes to improved mechanical properties, including higher M100. However, beyond this optimal loading, HNT agglomeration occurs, creating stress concentration points that compromise the material's structure. This agglomeration disrupts the uniform distribution of stress, reduces the effectiveness of stress transfer, and ultimately leads to a decrease in M100 [87, 88]. The M100 of SBR/XNBR/HNTs nanocomposites is 97% higher than that of the pure SBR/XNBR blends. The addition of HNTs significantly improves the CLD, acting as

crosslinking sites and enhancing the polymer network. This increased CLD results in a stiffer material with improved mechanical properties, including a higher M100. In contrast, SBR/XNBR blends show lower CLD values, leading to a less rigid material with lower M100. The higher CLD in the nanocomposites ensures a more tightly bound network of polymer chains, enhancing stress transfer and leading to a 97% greater increase in M100 compared to the blends without HNTs. The presence of HNTs overcomes the limitations of the SBR/XNBR blends by enhancing crosslinking and ensuring better stress distribution, leading to a significant improvement in M100 [89].

As depicted in Fig. 5, the tear strength (TAS) of SBR/XNBR/HNTs nanocomposites initially increases with the HNT content, peaking at 6 phr, before decreasing thereafter. Up to this optimal level, HNTs are well-dispersed within the SBR/XNBR matrix, effectively reinforcing the material. The elongated, rigid structure of HNTs enhances tear resistance by improving the composite's ability to transfer stress and resist the propagation of tears. Optimal HNT dispersion enhances the CLD, resulting in a stronger, more resilient network structure that hinders crack initiation and growth. At lower HNT content, the mobility of the polymer chains is slightly restricted, which contributes to improved toughness and tear resistance. However, beyond 6 phr, HNTs begin to agglomerate, forming clusters that create weak points, which compromise the composite's strength. Poor dispersion and clustering of HNTs diminish the uniformity of stress distribution, weakening the material and resulting in a decrease in TAS. Excessive HNTs can also interfere with crosslinking, reducing the effective CLD and compromising the polymer network's integrity, making it more prone to tearing [90–92]. The TAS of SBR/XNBR/HNTs nanocomposites shows a 91% increase compared to that of SBR/XNBR blends. The absence of HNTs in the blends results in a lower CLD, leading to a less robust network and greater susceptibility to tear propagation. Incorporating HNTs significantly improves the CLD, providing additional crosslinking sites and creating a tightly bound polymer network that enhances the material's tear strength. The improved network structure and higher CLD in the nanocomposites result in a more cohesive and resilient material, effectively resisting tear initiation and propagation. In contrast, the SBR/XNBR blends lack these reinforcing effects, resulting in lower TAS. The increased CLD in SBR/XNBR/HNTs nanocomposites facilitates more effective stress distribution across the matrix, reducing localized stress concentrations that can initiate tears. This enhancement in stress distribution leads to a more pronounced increase in TAS compared to the blends, where the lack of additional crosslinking leads to lower tear strength [93, 94]. Overall, the addition of

Fig. 5 The tear strength of SBR/XNBR/HNTs nanocomposites



HNTs provides a substantial improvement in tear resistance, with a 91% greater increase in TAS in the nanocomposites.

Figure 6 shows the hardness of SBR/XNBR/HNTs nanocomposites, which increases with higher HNT content. HNTs, rigid tubular structures, enhance the composite's rigidity by providing a stiffer framework within the SBR/XNBR matrix. Their inherent stiffness contributes to the hardness of the nanocomposite by reinforcing the material both physically and chemically. HNTs serve as additional crosslinking sites, increasing the CLD within the matrix. The higher CLD restricts the mobility of the polymer chains, resulting in a stiffer and harder material. The well-dispersed HNTs improve load distribution and efficiently transfer stress across the composite, reducing localized deformation and further increasing hardness. The strong bonding between HNTs and the SBR/XNBR matrix enhances the mechanical properties, creating a more cohesive structure that resists deformation under pressure [95–97]. The hardness of the SBR/XNBR/HNTs nanocomposites is 13% higher than that of the SBR/XNBR blends. The blends, with a lower CLD, are softer, as their polymer chains are more mobile and allow for greater deformation under pressure. In contrast, the addition of HNTs increases the CLD, stiffening the material and reducing its deformability. This results in increased resistance to indentation or penetration, correlating with higher hardness values. The higher CLD in the nanocomposites facilitates more efficient stress distribution, enabling the material to withstand higher loads without

significant deformation. The densely crosslinked structure provided by HNTs ensures that the composite maintains its shape and resists indentation or penetration forces. The robust interfacial bonding between the high-aspect-ratio HNTs and the SBR/XNBR matrix further enhances the hardness, thereby improving the material's overall mechanical properties [98, 99].

Rebound resilience of the SBR/XNBR/HNTs nanocomposites

Figure 7 demonstrates the rebound resilience (RR) of SBR/XNBR/HNTs nanocomposites, which decreases as HNT content increases. HNTs are rigid and have a higher modulus compared to the elastomeric SBR/XNBR matrix. As the content of HNTs increases, the stiffness of the nanocomposite becomes more prominent, limiting its ability to revert to its original shape after deformation. RR, which measures the ability to rebound after deformation, decreases as the material's stiffness increases. The incorporation of HNTs modifies the overall elastic modulus of the nanocomposite. While the elastomeric SBR/XNBR provides inherent elasticity, the stiffness of HNTs reduces the material's overall elasticity. Since resilience is tied to the ability to deform and return to the original shape, the increased stiffness from HNTs reduces RR. At higher HNT loadings, achieving uniform dispersion becomes challenging. Poor dispersion may cause agglomeration or clustering of HNTs, acting as

Fig. 6 The hardness of SBR/XNBR/HNTs nanocomposites

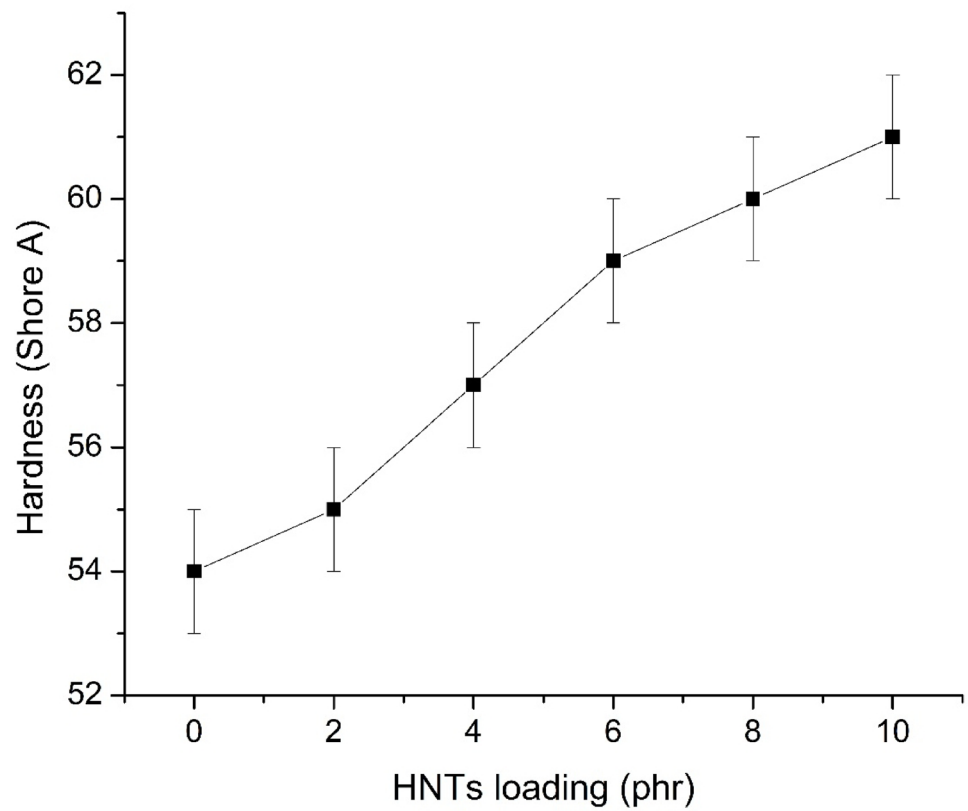
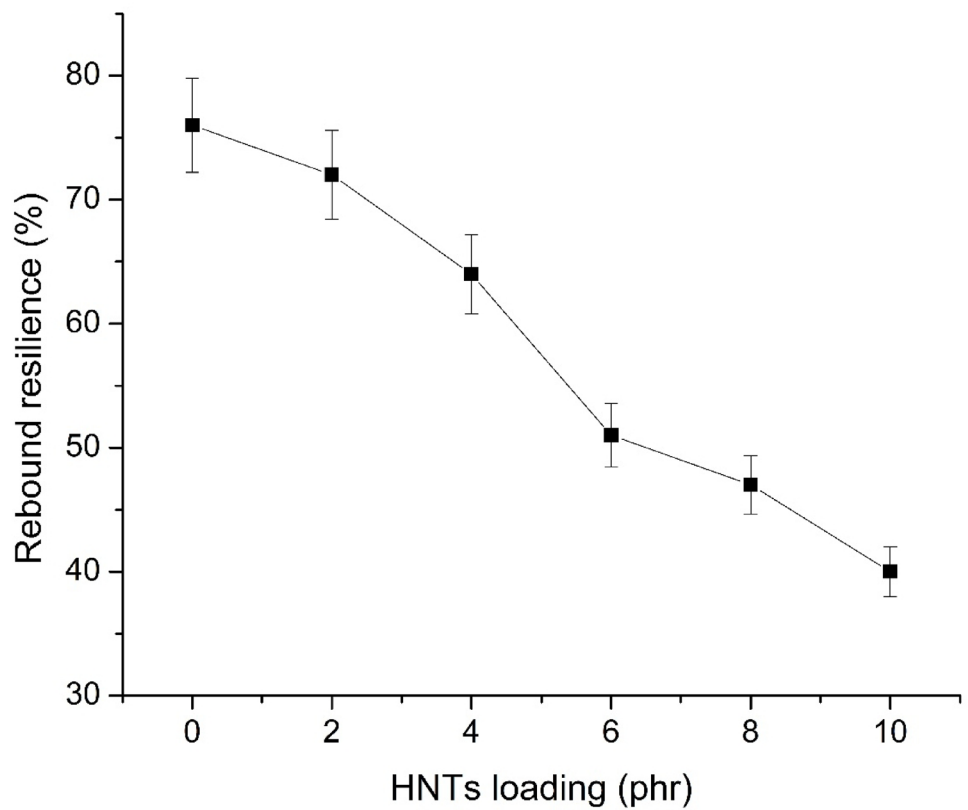


Fig. 7 The rebound resilience of SBR/XNBR/HNTs nanocomposites



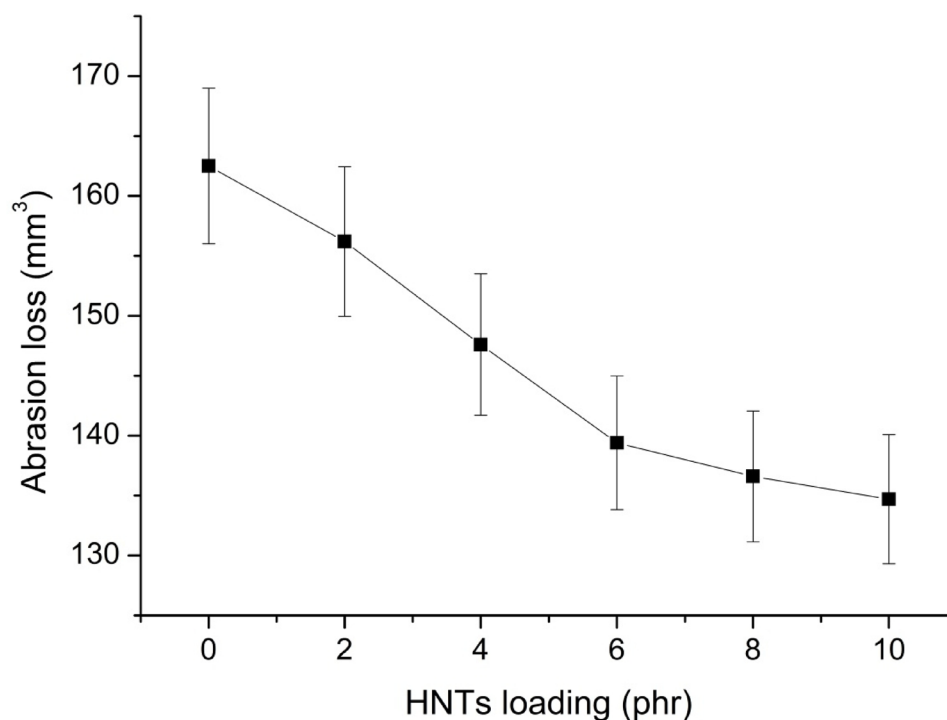
stress concentrators that compromise the nanocomposite's resilience. Non-uniform distribution of HNTs disrupts the composite's mechanical integrity, leading to reduced resilience. This effect becomes more significant as HNT content exceeds optimal levels. The damping characteristics of the nanocomposite can also be influenced by the dispersion and content of HNTs. Damping relates to energy dissipation during deformation, which impacts RR. There is an optimal balance where HNT addition enhances certain properties without significantly compromising RR, but exceeding this balance leads to a noticeable decline in resilience due to increased stiffness and altered damping behavior [100, 101]. The RR of SBR/XNBR/HNTs nanocomposites is 47% lower than that of the base blends. As the concentration of HNTs in the matrix increases, they function as fillers that enhance the overall stiffness of the composite, leading to a higher elastic modulus. Rubber materials typically exhibit high RR due to their ability to deform and recover quickly. However, as the content of HNTs increases, the material becomes less compliant and more resistant to deformation, leading to lower RR. Poor dispersion or aggregation of HNTs limits the movement of polymer chains within the SBR/XNBR matrix. The CLD, influenced by the presence of HNTs, impacts the mobility of polymer chains during deformation. When this mobility is restricted, the material's ability to return to its original shape is reduced, leading to a further decrease in RR. Changes in CLD due to HNT incorporation can alter the damping behavior, reducing the efficiency of energy storage and recovery, which is crucial for maintaining RR [102, 103]. The quality of the interaction

between HNTs and the SBR/XNBR matrix is crucial for RR. Poor dispersion or weak bonding between the HNTs and the matrix can result in stress concentrations and structural weaknesses, further diminishing the material's ability to recover its shape after deformation.

Abrasion resistance of the SBR/XNBR/HNTs nanocomposites

Figure 8 illustrates the abrasion loss of SBR/XNBR/HNTs nanocomposites, showing a clear decrease in abrasion loss with increasing HNTs content. This inverse relationship emphasizes the reinforcing effect of HNTs, which have a rigid, tubular structure and high aspect ratios. When incorporated into the SBR/XNBR matrix, HNTs enhance mechanical properties, particularly abrasion resistance, by reducing wear and tear through their inherent strength and rigidity. The addition of HNTs increases the CLD within the SBR/XNBR matrix, enhancing material integrity and cohesion. This densely crosslinked structure distributes mechanical stress more effectively, reducing material wear under friction. Dispersed HNTs form a protective layer on the surface, absorbing abrasive forces and reducing wear. As HNTs act as sacrificial barriers, they dissipate energy from abrasive impacts, lowering the overall wear rate. The strong interfacial bonding between HNTs and the polymer matrix enhances abrasion resistance by preventing filler pull-out or debonding under mechanical stress. This bonding helps maintain the composite's structural integrity in abrasive conditions. Proper dispersion of HNTs is crucial, as well-distributed

Fig. 8 The abrasion loss of SBR/XNBR/HNTs nanocomposites



fillers enhance resistance to abrasive particles by preventing surface penetration and material loss [104–106]. Higher HNTs content leads to increased CLD, creating additional bonding points between polymer chains and HNTs. This results in a harder, more cohesive material structure that withstands mechanical stress with minimal deformation [87]. The enhanced surface hardness, driven by HNTs and CLD, plays a vital role in reducing abrasion loss over time by resisting scratches and gouges. Overall, the combination of increased hardness, strong interfacial bonding, and improved crosslinking due to HNTs results in superior wear resistance and lower abrasion loss in SBR/XNBR nanocomposites.

Compression set of the SBR/XNBR/HNTs nanocomposites

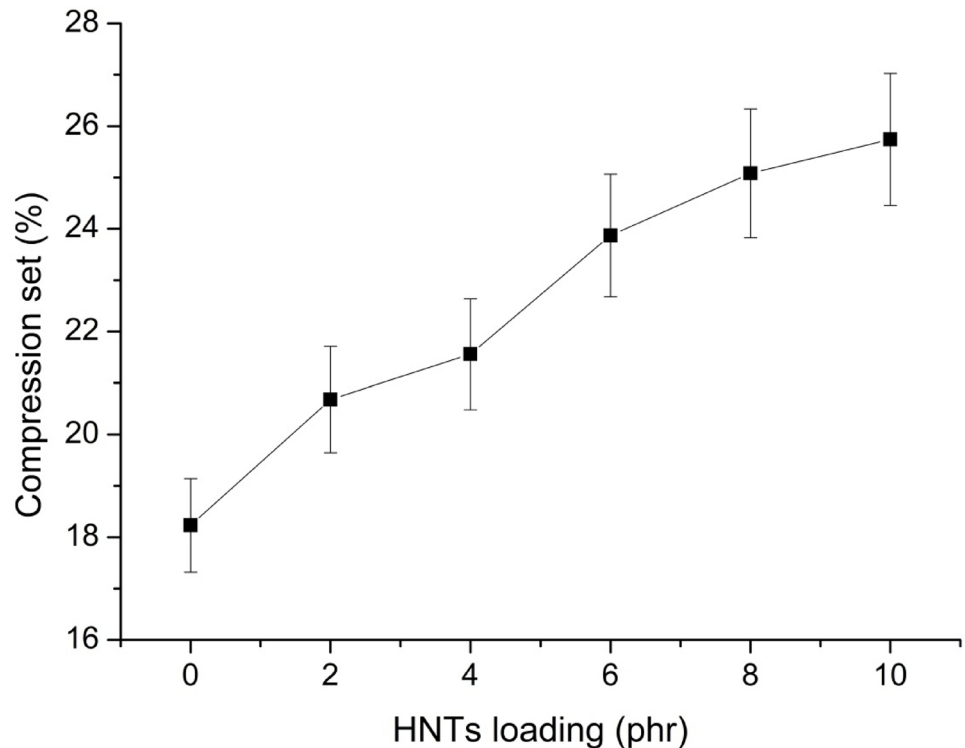
The compression set (CS) of SBR/XNBR/HNTs nanocomposites, shown in Fig. 9, increases with increasing HNT content. HNTs are rigid and have a higher modulus than the elastomeric SBR/XNBR matrix. As the HNT content increases, the stiffness contributed by these fillers becomes more pronounced, which reduces the material's ability to recover after deformation, leading to higher permanent deformation, as indicated by CS. SBR/XNBR elastomers are known for their high elasticity and ability to rebound after compression. However, an increased HNT content stiffens the composite, reducing its ability to fully regain its original shape. The movement and reformation of polymer chains are hindered by the increased HNT content, leading to greater

residual strain and higher CS. Moreover, the incorporation of HNTs can influence the CLD within the SBR/XNBR matrix. While a higher CLD, which is often promoted by HNTs, can improve mechanical properties, it also impacts elastic recovery. Enhanced crosslinking results in a more stable matrix but may increase the material's tendency to retain deformation over time, contributing to higher CS values. The quality of bonding between HNTs and the SBR/XNBR matrix is critical. Poor bonding or agglomeration of HNTs can lead to stress concentrations, weakening the nanocomposite and further exacerbating CS. As HNT content increases, the nanocomposite tends to exhibit higher hardness due to the rigid nature of HNTs and possibly increased CLD. A harder material is less compliant and more prone to permanent deformation under compression [107]. Consequently, the material's elastic deformation is restricted, leading to increased plastic deformation and a higher tendency for permanent set. In summary, as HNTs content increases, the nanocomposite's ability to recover after compression diminishes, leading to increased CS and reduced memory retention capacity. Over time, the material may become more susceptible to creep and relaxation under sustained loading conditions, further contributing to higher CS.

Crosslinking density of the SBR/XNBR/HNTs nanocomposites

Figure 10 presents the crosslink density (CLD) of SBR/XNBR/HNTs nanocomposites. Initially, the CLD increases

Fig. 9 The compression set of SBR/XNBR/HNTs nanocomposites



with the addition of HNTs up to 6 phr, beyond which it starts to decrease at higher loadings. HNTs are typically used as fillers in polymer matrices, and their addition to the SBR/XNBR matrix can promote crosslinking by offering more sites for crosslink formation. This leads to an increase in CLD up to an optimal loading of around 6 phr, where HNTs enhance polymer chain interactions and facilitate efficient crosslink formation [89]. At concentrations higher than 6 phr, the beneficial effects of HNTs may diminish, as excessive HNTs can agglomerate within the matrix, reducing the surface area available for crosslinking. This agglomeration hampers polymer-filler interactions and may decrease the overall CLD. The initial increase in CLD strengthens the nanocomposite, enhancing mechanical properties such as stiffness and TS. However, beyond the optimal HNTs loading, reduced polymer-filler interactions can weaken the polymer network and lower mechanical properties [108, 109]. CLD is crucial for improving TS, EB, M100, TAS, hardness, abrasion resistance, and CS. A higher CLD results in a more interconnected polymer network, improving resistance to tensile forces and leading to an increase in TS. A well-balanced CLD improves EB by distributing stress evenly and preventing premature chain slippage, which is critical in preventing early breakage under tension. Moreover, a denser crosslinked network improves the material's resistance to tear propagation, increasing TAS. It also enhances hardness by restricting polymer chain movement, which is vital for applications requiring indentation resistance. CLD increases abrasion resistance by

reducing material wear due to friction. Finally, higher CLD improves CS, ensuring dimensional stability and resilience in the material over time.

Swelling resistance of the SBR/XNBR/HNTs nanocomposites

Effect of nanofillers

Table 2 displays the mole percent uptake (MPU) of various solvents in SBR/XNBR/HNTs nanocomposites at 30 °C. MPU initially decreases with increasing HNTs content up to 6 phr, but then rises at higher HNT loadings. At lower concentrations of HNTs (up to 6 phr), there is generally better dispersion within the SBR/XNBR matrix. Properly dispersed fillers provide a larger surface area for interaction with the polymer chains, resulting in better encapsulation and bonding. With increasing HNTs content, each nanotube effectively occupies more available surface area in the SBR/XNBR matrix. This results in a more efficient use of the available polymer-filler interaction sites, which in turn lowers the mole percent uptake per unit volume of the composite. Beyond 6 phr, the potential for filler agglomeration grows. Agglomerated fillers reduce the effective surface area for interaction with the SBR/XNBR matrix, causing a decrease in the overall MPU. As the HNT concentration rises further, there may come a point where the filler network begins to form interconnected clusters or agglomerates. This further reduces the

Fig. 10 The crosslinking density of SBR/XNBR/HNTs nanocomposites

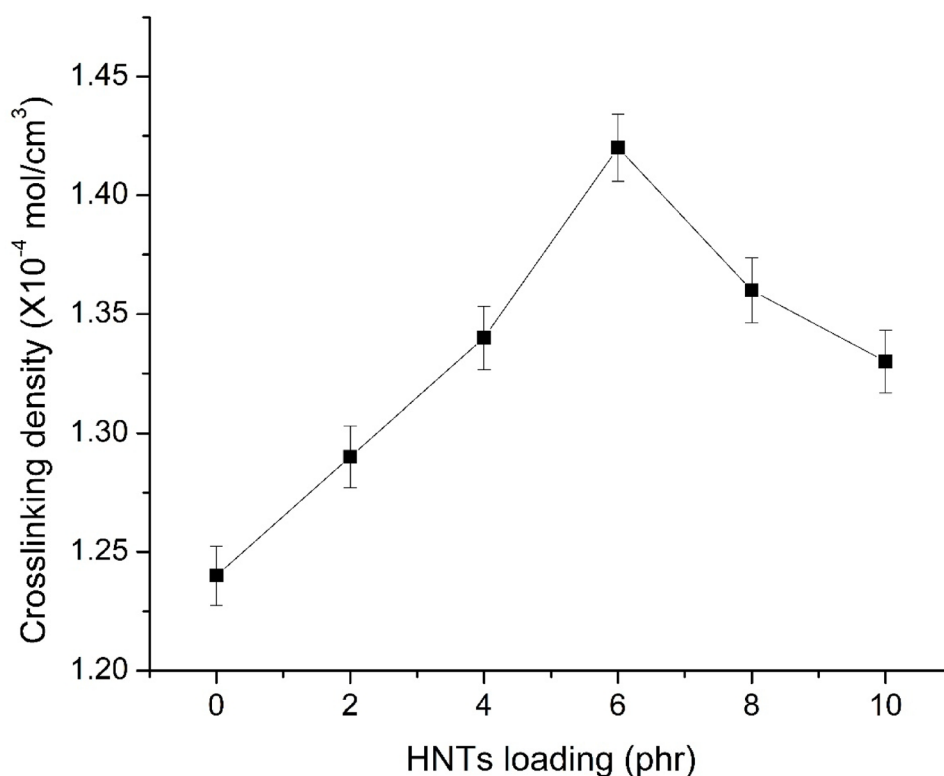


Table 2 Mole percent uptake of SBR/XNBR-HNTs nanocomposites at 30 °C

Sample code	Mole percent uptake (mol%)										
	Aromatic solvents				Aliphatic solvents				Chlorinated solvents		
	Benzene	Toluene	Xylene	Mesitylene	n-pentane	n-hexane	n-heptane	n-octane	Dichloromethane	Chloroform	Carbon tetrachloride
H ₀	2.68	2.45	2.31	2.07	0.45	0.27	0.23	0.17	2.48	2.27	2.12
H ₂	2.61	2.38	2.26	2.02	0.42	0.24	0.21	0.15	2.43	2.24	2.08
H ₄	2.54	2.32	2.22	1.97	0.4	0.22	0.19	0.12	2.4	2.19	2.05
H ₆	2.46	2.23	2.13	1.89	0.37	0.17	0.15	0.1	2.32	2.13	2.01
H ₈	2.49	2.28	2.16	1.92	0.39	0.19	0.17	0.12	2.36	2.16	2.06
H ₁₀	2.52	2.31	2.19	1.96	0.41	0.21	0.2	0.14	2.39	2.19	2.09

effective dispersion and accessible surface area for polymer interaction, thereby decreasing the MPU. This results in a more efficient use of the available polymer-filler interaction sites, which in turn lowers the mole percent uptake per unit volume of the composite. Beyond 6 phr, the potential for filler agglomeration grows. Agglomerated fillers reduce the effective surface area for interaction with the SBR/XNBR matrix, causing a decrease in the overall MPU. As the HNT concentration rises further, there may come a point where the filler network begins to form interconnected clusters or agglomerates [110, 111]. Efficient filler dispersion also impacts barrier properties such as gas permeability and moisture resistance, which are crucial in various applications.

Effect of solvents

The order of uptake of a material in different aromatic solvents such as mesitylene, xylene, toluene, and benzene depends on several factors related to the solvents' chemical structure and physical properties: Aromatic solvents exhibit different levels of polarity, which arise from variations in their chemical structures [112]. Benzene has lower polarity compared to toluene, xylene, and mesitylene, with the latter solvents showing increased polarity due to the presence of methyl groups. Generally, solvents with higher polarity can interact more strongly with solutes through dipole–dipole interactions or hydrogen bonding. This stronger interaction can facilitate higher uptake of solutes. The size and structure of the aromatic solvents can affect the accessibility of the solute molecules to the solvent molecules. Larger solvents may have more restricted access points for solute uptake compared to smaller solvents. Smaller aromatic solvents like benzene may allow for easier diffusion and permeation of solutes due to their smaller molecular size and potentially less hindered pathways. The solubility parameter and compatibility between the solute and solvent play a crucial role. Aromatic solvents with higher solubility parameters and better compatibility with the solute molecule can lead to higher uptake [113]. Miscibility and solubility of the solute in the solvent

also influence the uptake. Solvents where the solute is more soluble tend to have higher uptake capacities. Experimental conditions such as temperature and pressure can influence the solvation process and affect the uptake capacity of solvents. The method used for measuring uptake, such as static or dynamic techniques, can impact the observed order of uptake. Dynamic methods, for example, may allow for better mixing and higher uptake rates compared to static methods.

The sequence of uptake of a material in aliphatic solvents like n-pentane, n-hexane, n-heptane, and n-octane can be attributed to several key factors, including the solvent's chemical structure, molecular size, and its interaction with the solute: Aliphatic solvents differ primarily in their carbon chain length [114]. n-Pentane has the shortest carbon chain, followed by n-hexane, n-heptane, and n-octane with progressively longer chains. Longer carbon chains in solvents like n-heptane and n-octane can hinder the solute molecules from accessing solvent molecules, potentially reducing the uptake capacity compared to shorter chain solvents like n-pentane and n-hexane. Aliphatic solvents primarily interact through London dispersion forces, which increase with the size of the solvent molecule. Larger molecules such as n-heptane and n-octane can exhibit stronger dispersion forces, potentially leading to higher solute uptake. Generally, aliphatic solvents are less polar compared to aromatic solvents. The slight differences in polarity between n-pentane, n-hexane, n-heptane, and n-octane may influence their solvation capabilities and solute uptake. Solvents with similar solubility parameters to the solute molecule are more likely to dissolve and solvate the solute effectively, leading to higher uptake. The miscibility of the solute in the solvent also plays a role [114]. Solvents where the solute is more miscible tend to have higher uptake capacities due to better solvation. Experimental conditions such as temperature and pressure can affect solubility and solvation kinetics, thereby influencing the observed order of uptake. The method used for measuring uptake (e.g., static or dynamic techniques) can impact results. Dynamic methods may enhance mixing and improve solute–solvent contact, potentially increasing uptake rates.

The sequence of uptake of a material in chlorinated solvents like dichloromethane, chloroform, and carbon tetrachloride can be explained by several important factors, including the solvents' chemical structure, polarity, and their interactions with the solute: Dichloromethane is more polar compared to chloroform and carbon tetrachloride. Polar solvents generally exhibit stronger interactions with polar or ionic solutes through dipole–dipole interactions or hydrogen bonding, which can enhance solute uptake [114]. While all three solvents primarily interact through London dispersion forces, dichloromethane's higher polarity can enhance solvation and increase solute uptake compared to less polar solvents like chloroform and carbon tetrachloride. The solubility parameter of a solvent influences its ability to dissolve and solvate solutes effectively. Solvents with closer solubility parameters to the solute molecule are typically more effective in solute uptake. The extent of solute miscibility in the solvent also affects uptake. Solvents where the solute is more miscible tend to have higher uptake capacities due to better solvation and dissolution. Carbon tetrachloride has a larger molecular size and less electronegative atoms compared to dichloromethane and chloroform. This difference in molecular structure affects solvent–solute interactions and can influence uptake capacity [114]. Larger molecules may have more restricted access points for solute molecules, potentially reducing uptake capacity compared to smaller, more accessible molecules like dichloromethane and chloroform. The toxicity and environmental impact of chlorinated solvents vary. Dichloromethane and chloroform are commonly used in various industrial applications but require careful handling due to their potential health risks. Carbon tetrachloride is less commonly used today due to its higher toxicity and environmental concerns.

The order of uptake in aromatic solvents, from mesitylene to benzene, reflects a combination of solvent properties such as polarity, size, solubility, and compatibility with the solute. In aliphatic solvents, the uptake increases from n-octane to n-pentane and is influenced by factors such as molecular size, intermolecular forces, solubility parameters, and experimental conditions. In chlorinated solvents, the uptake decreases from dichloromethane to carbon tetrachloride, and this is determined by solvent polarity, solubility characteristics, molecular structure, and safety considerations.

Morphology of the SBR/XNBR/HNTs nanocomposites

The FESEM micrographs shown in Fig. 11 illustrate the tensile-fractured surfaces of SBR/XNBR composites containing different HNT loadings: (a) 0 phr, (b) 2 phr, (c) 4 phr, (d) 6 phr, (e) 8 phr, and (f) 10 phr. Figure 11a shows the neat SBR/XNBR blend, revealing a smooth and homogeneous surface without visible phase separation, indicating good compatibility between the SBR and XNBR phases. With increasing

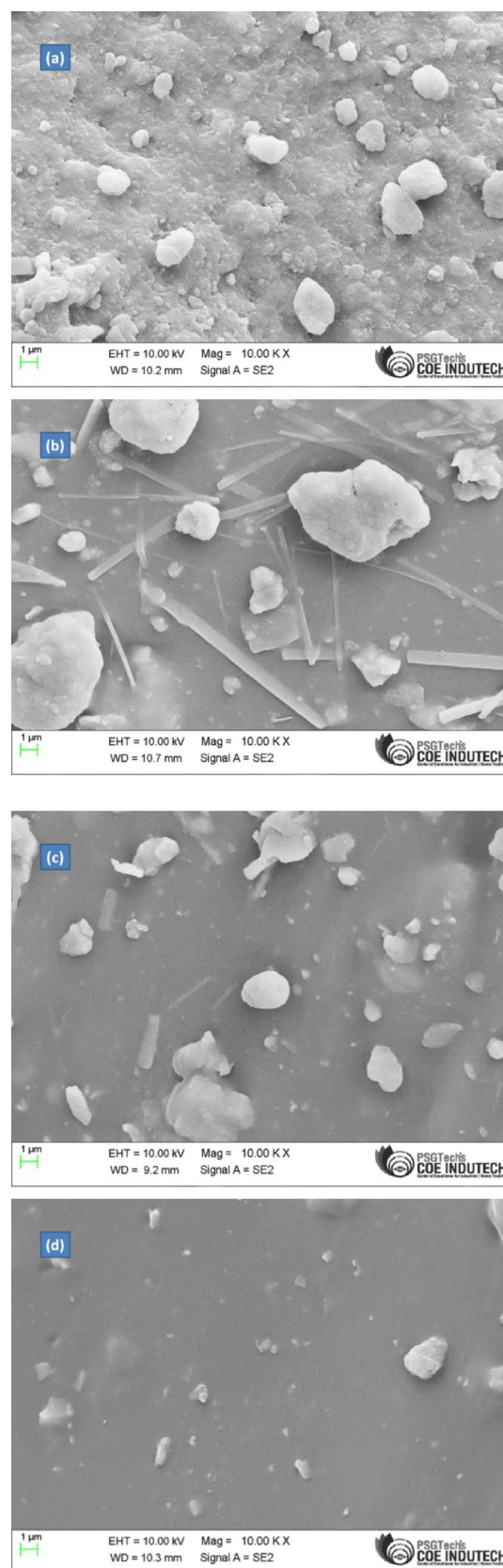


Fig. 11 FESEM images of 0-HNTs (a), 2-HNTs (b), 4-HNTs (c), 6-HNTs (d), 8-HNTs (e) and 10-HNTs (f) in the SBR/XNBR rubber matrix

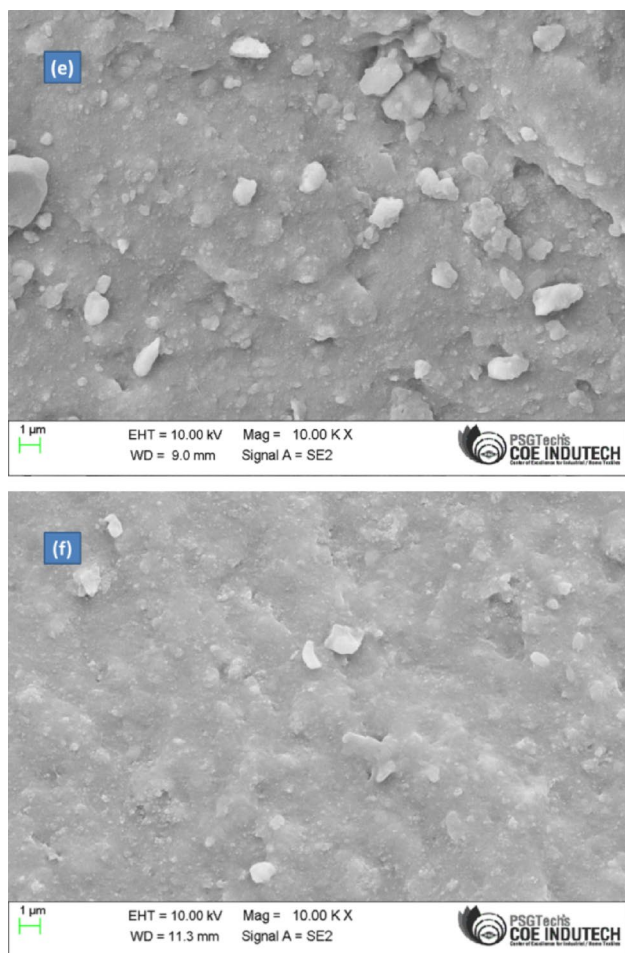


Fig. 11 (continued)

HNT content from 2 to 6 phr [Fig. 11b–d], the micrographs show well-dispersed HNTs embedded uniformly within the matrix. The needle-like morphology and high aspect ratio of HNTs are particularly visible in Fig. 11b and c, where individual nanotubes are separated and evenly distributed, promoting efficient stress transfer and enhanced toughness.

At 8 phr HNTs [Fig. 11e], slight agglomeration is observed. This is evident from localized clusters of densely packed HNTs and microparticle domains on the fractured surface. These clusters reduce the effective filler–matrix contact area and create stress concentration sites, limiting reinforcement efficiency. At 10 phr [Fig. 11f], more pronounced clustering occurs, where HNTs aggregate into larger domains, forming visibly overlapping structures that disrupt the continuity of the polymer–filler network. This aggregation is accompanied by less roughness in the surrounding matrix, indicating reduced energy dissipation and filler integration. The progression from uniform dispersion (≤ 6 phr) to noticeable clustering at higher loadings confirms the existence of an optimal filler concentration, beyond which filler–filler interactions

dominate over filler–matrix interactions, leading to diminished mechanical performance [115–117].

Conclusions

The SBR/XNBR nanocomposites, carefully prepared with HNTs using an open mill mixer for thorough dispersion and followed by a sulfur vulcanization process, have provided valuable insights, resulting in the following conclusions from the study:

1. The increase in torques (M_L , M_H , and ΔM) and CRI indicates an enhancement in crosslink density and improved vulcanization efficiency due to the incorporation of HNTs in the SBR/XNBR nanocomposites. Additionally, the decrease in t_{s2} and t_{90} suggests accelerated curing kinetics, attributed to the nucleating and catalytic effects of HNTs during vulcanization.
2. The TS, EB, M100, and TAS of the SBR/XNBR nanocomposites improve as HNT content increases up to 6 phr. However, they begin to decline with further HNT loading. This behavior reflects the optimal reinforcement effect of HNTs at lower concentrations, which enhances mechanical properties. Beyond this point, excessive filler loading disrupts the SBR/XNBR matrix, reducing performance.
3. The percentage increase in TS of the SBR/XNBR nanocomposites is approximately 67%, EB by 50%, M100 by 75%, and TAS by about 12% when compared to the pure SBR/XNBR blend. These improvements are attributed to the higher crosslink density induced by the nanocomposites, which strengthens intermolecular interactions and reinforces the polymer structure.
4. The hardness and abrasion resistance of the SBR/XNBR composites increase with HNT content, while RR decreases. This trend results from the stiffening effect of HNTs, which enhance material hardness and wear resistance but diminish the material's ability to recover energy after deformation.
5. The CS of the SBR/XNBR composites rises with increased HNT content. This is because the presence of HNTs restricts polymer chain mobility, leading to less efficient recovery from compression deformation.
6. The swelling resistance of the SBR/XNBR composites increases with the addition of HNTs content up to 6 phr and then declines with further loading. Initially, HNTs enhance filler–matrix interactions and reduce polymer permeability, improving swelling resistance. However, beyond 6 phr, excessive HNTs loading may create agglomerates or disrupt polymer structure, compromising the composite's ability to resist swelling.

7. The HNTs-reinforced SBR/XNBR composite has potential applications in high-performance automotive tires, industrial seals, and gaskets due to its enhanced TS, TAS, and abrasion resistance. These findings could drive the rubber industry towards more durable, eco-friendly alternatives. To adopt these novel compositions, tire manufacturers should conduct extensive testing for real-world performance, develop scalable production techniques, and ensure compatibility with existing tire manufacturing processes.

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Data availability The datasets used and/or analyzed in this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors confirm that there are no known financial or personal relationships that could have influenced the research presented in this paper.

Ethical approval This research, which involved only non-human subjects, was approved by the Department of Mechanical Engineering at EGS Pillai Engineering College in Nagapattinam and SA Engineering College in Chennai. As no human or animal participants were involved, informed consent was not applicable.

Consent for publication As the corresponding author, I, Dr. V. Sivaramakrishnan, affirm that the manuscript titled "Reinforcement Potential of HNTs in SBR/XNBR Blend Nanocomposites" presents original research that has not been previously published or submitted for publication elsewhere. All co-authors have made substantial contributions to the work, approved the content, and are aware of this submission. The manuscript does not violate any copyright and complies with the journal's guidelines. I consent to the journal's peer review process.

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